Definitions

1 Machine Learning

1.1 inductive bias

- Restricting the hypothesis space $\mathcal{H}$ to a certain set of hypotheses.

  Smaller $\mathcal{H}$ means less data needed to pin down the right hypothesis h.

1.2 example: symmetry as inductive bias

A CNN restricts $\mathcal{H}$ from all functions $\mathbb{R}^{d\times d} \to \{0,1\}$ to just the translation-equivariant ones,

$$\mathcal{H} = \{f : f(\tau_a x) = \tau_a\, f(x) \quad \forall a\},$$

where τ_a translates the image by a. The network never has to spend training data learning that an edge is an edge at every position, since the hypothesis cannot express "not an edge at position 37" — that is why it generalizes.